

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

**Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)**

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

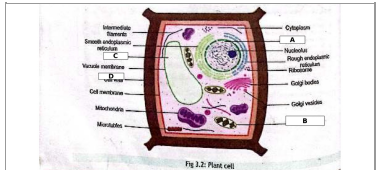
1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number of questions, choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Who is regarded as the first person to classify living organisms?	John Ray	Tournefort	Aristotle	Carl Woese	☐
2	Which scientist introduced the three-kingdom system, adding kingdom Protista?	Robert Whittaker	Ernst Haeckel	Carl Woese	Carolus Linnaeus	☐
3	Each centriole consists of how many triplets of microtubules?	Two	Three	Nine	Twenty	☐
4	In tissue culture, the agar growth medium contains nutrients and which plant hormone?	Gibberellins	Cytokinins	Ethylene	Auxins	☐
5	Which organ system regulates blood calcium by storing/releasing it from bones?	Muscular	Endocrine	Skeletal	Urinary	☐
6	Who first isolated nucleic acid material (as "nuclein") from human pus cells?	James Watson	Robert Hooke	Francis Crick	Friedrich Miescher	☐
7	The diameter of the DNA double helix is uniformly about:	0.2 nm	2 nm	20 nm	200 nm	☐
8	Human enzymes generally function best within which temperature range?	0-10°C	20-25°C	36-38°C	50-60°C	☐
9	Trypsin (in the small intestine) performs best at an optimum pH of about:	2.00	7.00	8.00	10.00	☐
10	Complete oxidation of one glucose molecule yields approximately how many net ATP?	2	4	18	36	☐
11	Collected data is organized using tables/graphs, then analyzed by means of:	percentages only	ratio and proportion	observation alone	hypothesis testing	☐
12	About how many types of specialized cells arise from a single zygote in the human body?	50	120	220	400	☐

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	State the THREE criteria that Carl Woese used to classify organisms into the three domains of life (Archaea, Bacteria and Eukarya).	3	OR	In the two-kingdom classification system (the oldest classification scheme), name the two kingdoms used. State which groups of organisms (besides autotrophs) were included in the kingdom Plantae, and which organisms were placed in kingdom Animalia.	3
2(ii)	Study the figure below, which shows a plant cell (Fig 3.2, adapted from your textbook). Identify the structures labelled A, B, C, and D. 	3	OR	Define cell specialization (differentiation). Describe the structure of a root hair cell (a modified epidermal cell of a plant) and explain how this structure helps it perform its function of absorbing water and minerals from the soil.	3
2(iii)	What is a haematopoietic stem cell? Name any FOUR types of cells that haematopoietic stem cells can give rise to, according to the textbook's diagram.	3	OR	Describe the THREE types of vacuoles mentioned in the textbook: the temporary vacuoles of animal cells, the contractile vacuoles of freshwater organisms (e.g. Amoeba), and the large central vacuole of plant cells. State ONE function of	3

				each.	
2(iv)	According to the textbook's STEAM Activity for studying a plant cell, describe the steps used to prepare an onion skin slide, from placing the skin in water to observing it under the microscope.	3	OR	According to the textbook's STEAM Activity for studying an animal cell, describe the steps used to prepare a human cheek cell slide, including the stain used, from sample collection to observing it under the microscope.	3
2(v)	State the diploid and haploid chromosome numbers in human cells. Using these numbers, explain why meiosis (and not mitosis) must occur during the formation of gametes, to maintain a constant chromosome number across generations.	3	OR	Describe the main events that occur during the G1 phase of interphase. Why must a cell form enzymes and nucleotides during this phase before DNA replication begins in the S phase?	3
2(vi)	Name the organs included in the skeletal system of the human body, and state its role in maintaining homeostasis, according to the textbook.	3	OR	Name the organs included in the urinary system of the human body, and state its role in maintaining homeostasis, according to the textbook.	3
2(vii)	Who first isolated nucleic acid material from the nucleus of human pus cells, and what did he name this material? Explain how the name "nucleic acid" came about.	3	OR	Describe the iodine test for starch (procedure and colour change observed). How would the result differ if the same test were performed on a sample containing glycogen instead of starch?	3
2(viii)	Describe the structure of the RNA molecule. Is it single or double stranded? Name its pentose sugar and the nitrogenous base that replaces thymine. Why are the nitrogen bases of RNA described as "overhung"?	3	OR	Monosaccharides such as ribose can exist in an open chain structure or a ring structure. Explain when each form occurs, and state which structure is more commonly found in the nucleotides of nucleic acids.	3
2(ix)	State the optimum pH for pepsin (in the stomach) and trypsin (in the small intestine). Explain why these two enzymes need such different optimum pH values, given where each works in the digestive system.	3	OR	Explain why very low temperatures make enzymes inactive rather than permanently denaturing them, and why enzyme activity is regained when normal temperature is restored. Contrast this with the effect of very high temperatures on enzyme structure.	3
2(x)	State how many ATP molecules are produced from the complete oxidation of one glucose molecule in cellular respiration, and how many moles of ATP an average human cell uses per day, according to the textbook.	3	OR	In the textbook's STEAM Activity, starch solution with amylase enzyme was tested at five temperatures (0°C, 20°C, 40°C, 60°C, 80°C) and then checked with iodine solution. At 40°C no blue-black colour appeared, but at 0°C, 60°C and 80°C it did. Explain why the result differed at these temperatures.	3
2(xi)	With reference to Fig 7.2 of the textbook, explain how the enzyme mechanism differs when catalyzing a SYNTHESIS reaction (combining two separate substrates into one product) compared to a BREAKDOWN reaction (splitting one substrate into two products). Name the two complexes formed in each case.	3	OR	According to the "Science Tidbits" box in the textbook, explain how chewing food for longer helps digestion: describe its effect on amylase production and on the activation of the enzyme pepsin in the stomach.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	Draw a well-labelled diagram of a neuron (nerve cell), showing its cell body, dendrites, and axon. State ONE function of each of these three parts.	5	OR	Describe how the structure of a neuron (cell body, dendrites, axon) is suited to its function of coordination in the body. Contrast this with the structure of a liver cell (round, prominent nucleus, abundant organelles) and a mesophyll cell (containing many chloroplasts), explaining how each cell's distinct structure relates to its specific function.	5
3(ii)	Describe the plant's two organ systems: the shoot system and the root system. Include the vegetative and reproductive sub-divisions of the shoot system, and state TWO functions of each system, referring to the textbook's Fig 5.6.	5	OR	Using a comparison table, describe any FOUR organ systems of the human body (choosing from Skeletal, Urinary, Lymphatic, Muscular, Immune, Nervous), stating for each system its main organs and its specific role in homeostasis, according to the textbook's Table 5.2.	5
3(iii)	In a double-stranded DNA molecule, 34% of the nitrogen bases are cytosine. Calculate the percentage of guanine, adenine, and thymine in this molecule, explaining your reasoning using the base-pairing rule of the Watson-Crick model.	5	OR	Describe THREE protein functions from the textbook (other than enzymes, structural proteins like collagen/keratin, or transport/immune proteins): (a) cell membrane proteins acting as channel/carrier proteins and as antigens defining blood groups, (b) fibrinogen in blood clotting, and (c) hormones as chemical messengers, naming TWO processes they regulate.	5
3(iv)	Study Fig 7.3 (effect of temperature), Fig 7.4 (effect of pH on pepsin and trypsin), and Fig 7.5 (effect of substrate concentration) from the textbook. For EACH graph, describe what happens to the rate of enzyme reaction as the x-axis variable increases, and name the phenomenon responsible when the rate stops increasing or starts to decrease.	5	OR	In the textbook's STEAM Activity 7.2, egg albumin protein solution was treated with pepsin enzyme at three different pH levels (neutral in tube A, acidic in tube B, alkaline in tube C) and then tested with Biuret solution. Describe the expected result (colour change) in each tube, and explain the reason in terms of pepsin's optimum pH and enzyme denaturation.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

— End of Question Paper —